

Business & Technical Justification: Enterprise Access to Claude.ai & RDF Studio for Knowledge-Driven Analytics

Program: Knowledge-Driven Analytics (KDA) Capability

Pilot Deliverable: HR Advisory 360 Semantic Intelligence Agent (Design-Time POC)

Sponsoring Organization: Enterprise Analytics

Key Stakeholders: IT Architecture & Review Board, HR Technology Steering Committee, Enterprise Tooling Procurement

Requested Tooling:

1. **Claude.ai** (Enterprise / Team Workspace) — Semantic reasoning, mapping synthesis, and impact-simulation workbench.
2. **RDF Studio** — Visual ontology/taxonomy modeling environment, SKOS scheme design, and schema governance hub.

1. Executive Summary

Enterprise Analytics proposes establishing a **Knowledge-Driven Analytics (KDA)** design-time proof of capability.

Modern enterprise data initiatives struggle when business meaning is buried in disconnected operational tables and brittle pipeline transformations. Knowledge is fundamentally a business-owned asset. Under the KDA paradigm, Enterprise Analytics formalizes business definitions, policies, and operational taxonomies into semantic models that empower business stakeholders while providing IT with stable, resilient data consumption layers.

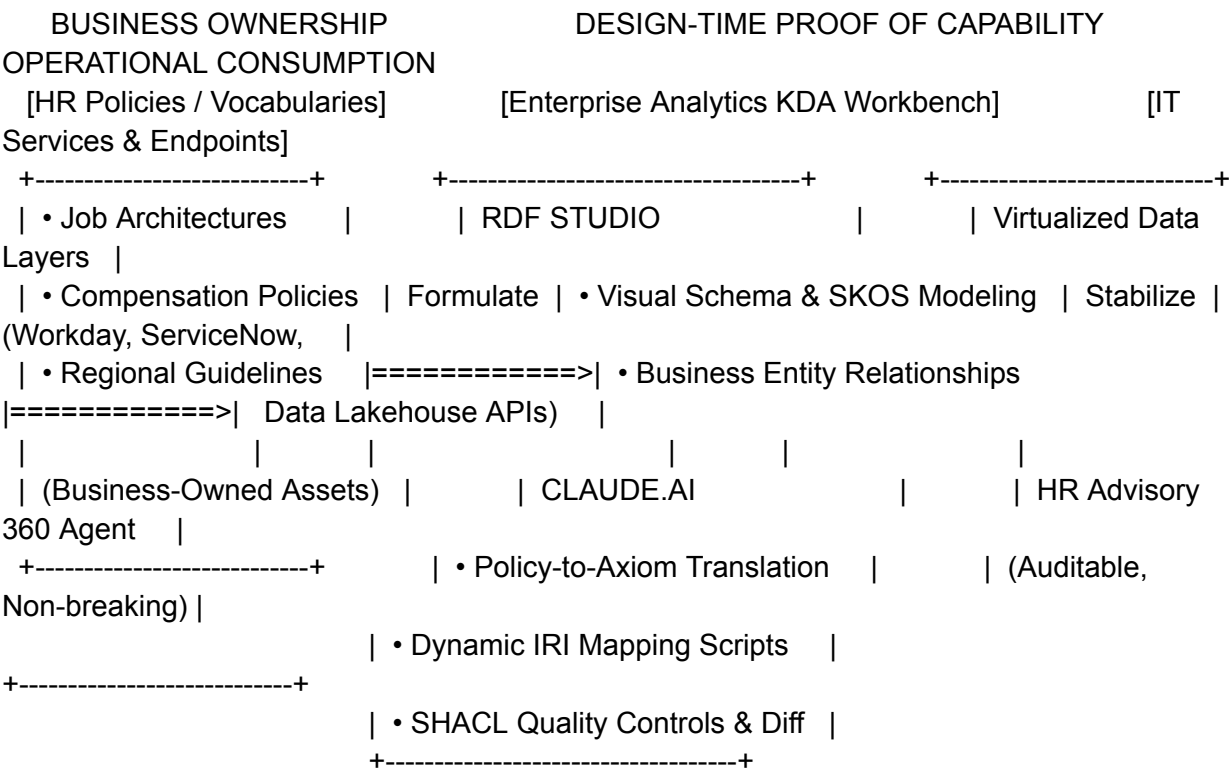
To establish this capability, Enterprise Analytics requires a combined design-time workbench:

- **RDF Studio:** The visual semantic modeling and governance workbench used to author, visualize, and maintain enterprise taxonomies, domain schemas, and canonical graphs.
- **Claude.ai:** The generative semantic engine used by analytics modelers to draft SKOS concept hierarchies, automate cross-schema IRI mappings, author data validation shapes (SHACL), and simulate change impact before deployment.

The proving ground for this tooling stack is the **HR Advisory 360 Agent**, an analytics-driven assistant designed to answer multi-jurisdiction, policy-constrained workforce questions that traditional Business Intelligence (BI) dashboards and vector search (RAG) cannot resolve.

2. Strategic Context: Why Knowledge-Driven Analytics?

Enterprise Analytics has established trusted partnerships with business leadership across domains. Moving from static tabular reporting to Knowledge-Driven Analytics allows the business to own its business rules and terminology directly, rather than relying on IT to repeatedly hard-code logic into one-off data pipelines.



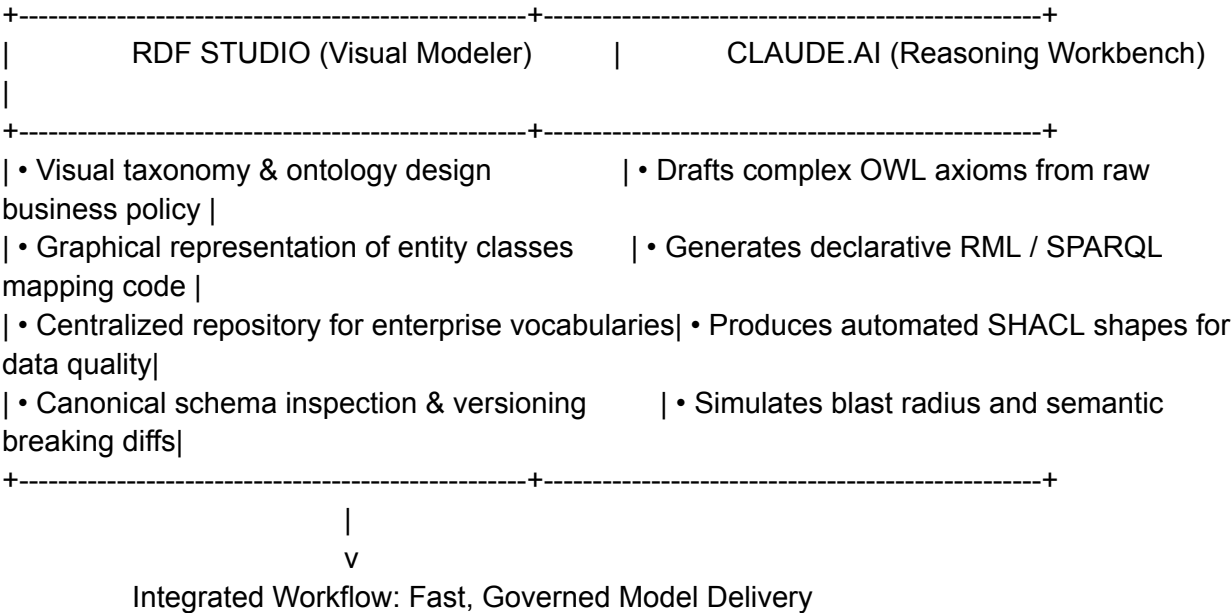
Business Impact: HR Advisory 360 Pilot

Operational Dimension	Current Analytics / IT State	KDA Approach (RDF Studio + Claude.ai)	Business Outcome
Cross-System Analysis	Manual reconciliation across Workday, ServiceNow, and regional policy PDF repositories.	Unified semantic layer connecting employee profiles to business rules and job frameworks.	70% reduction in time-to-insight for complex HR policy inquiries.

Schema Drift & Fragility	Vendor schema changes break ETL pipelines and invalidate operational dashboards.	Virtualized semantic mappings insulate analytics products from source schema shifts.	Elimination of pipeline breaks during quarterly SaaS upgrades.
Compliance & Traceability	Black-box analytics; difficulty proving why a specific policy path was recommended.	Explicit semantic graphs with full provenance (PROV-O) tracking rule execution.	Audit-ready lineage for compliance and internal controls.

3. The Dual Tooling Stack: Complementary Roles

RDF Studio and Claude.ai fulfill distinct, non-overlapping functions within the Enterprise Analytics design workflow.



1. **RDF Studio (The Canvas):** Provides the visual, human-friendly environment where analysts and business domain stewards collaborate to structure the HR domain: jobs, locations, benefit structures, and compliance classes.
2. **Claude.ai (The Co-Pilot):** Eliminates syntax friction and logic bottlenecks. It converts messy policy manuals into precise SKOS concepts, drafts RML/SPARQL mappings between operational table columns and model IRIs, and creates rigorous SHACL constraints to prevent bad data from reaching the model.

4. Technical Workflows for the HR 360 Pilot

4.1 Business Vocabulary & Taxonomy Harmonization

- Enterprise Analytics utilizes **RDF Studio** to establish standard SKOS-XL concept schemes for corporate leveling, roles, and competencies.
- **Claude.ai** ingests external industry frameworks (e.g., O*NET, standard labor codes) and corporate job descriptions, generating alignment suggestions and hierarchy candidates directly importable into RDF Studio.

4.2 Building the Virtualized Insulation Layer

- Operational HR systems (Workday, ServiceNow) expose data via REST endpoints and SQL views.
- Rather than loading everything into another physical data warehouse, Enterprise Analytics designs a **virtualized graph layer**.
- Using Claude.ai, analysts generate declarative schema-to-ontology mapping definitions (RML, SPARQL Construct). This mapping acts as an abstraction layer: if an underlying field name changes in Workday, analysts update the mapping in minutes without requiring IT to refactor downstream reporting or agent APIs.

4.3 Automated Semantic Quality Assurance & Metrology

- Quality governance is established upstream before data reaches consumers.
- Analysts prompt Claude.ai with business constraints (e.g., *"Senior Advisory roles require Grade 12+ and jurisdictional licensing in California"*).
- Claude compiles these rules into **SHACL (Shapes Constraint Language)** definitions validated inside RDF Studio, providing automated validation and error logging for every test scenario.
- **Impact Simulation:** When HR changes a sabbatical or retention policy, Claude simulates the changes against baseline queries to identify potential unintended downstream side effects before production sign-off.

5. Value to IT Services: De-risking & Offloading Technical Debt

This request directly aligns with IT Services' strategic priorities:

- **Offloading Domain Maintenance:** IT is relieved from maintaining complex, ever-shifting business logic inside brittle ETL/ELT pipelines. Enterprise Analytics governs the semantic layer in partnership with business owners.
- **Stable Consumption Endpoints:** IT receives stable, IRI-grounded semantic endpoints to feed downstream applications and microservices, insulated from backend SaaS schema churn.
- **No Direct Production Exposure:** Claude.ai and RDF Studio are strictly **design-time development tools**. They operate in analytical staging and modeling sandboxes. No

automated generative code or changes reach production systems without passing standard CI/CD regression tests and Git-based peer reviews.

- **Enterprise Security Standards:**

- Claude.ai Enterprise Tier ensures customer inputs and outputs are never utilized for model training.
- Role-based access control (RBAC) managed via corporate Okta/SAML SSO.
- Work is conducted strictly on conceptual schemas, metadata, and anonymized synthetic test sets—production PII/PHI is completely excluded from the modeling environment.

6. Pilot Implementation & Resource Request

To establish this design-time proof of capability for the HR Advisory 360 pilot, Enterprise Analytics requests approval for a **90-day evaluation window**:

Milestone	Timeframe	Deliverables
Phase 1: Environment & Setup	Weeks 1–3	Provision 3–5 seats of Claude.ai (Enterprise Tier) and RDF Studio; configure SSO and standard prefixes.
Phase 2: HR 360 Semantic Modeling	Weeks 4–7	Build core HR taxonomy in RDF Studio; use Claude.ai to draft virtualized schema mappings and policy axioms.
Phase 3: Validation & Agent Testing	Weeks 8–10	Apply SHACL validation suites; run semantic drift simulations against mock HR advisory queries.
Phase 4: Readout & Architecture Review	Weeks 11–12	Present proof-of-capability results to IT Services and HR Steering Committee, demonstrating pipeline stability and analytical precision.